EMS612U Flight dynamics and control
Tutorial 2 typed solutions

2.1

Data

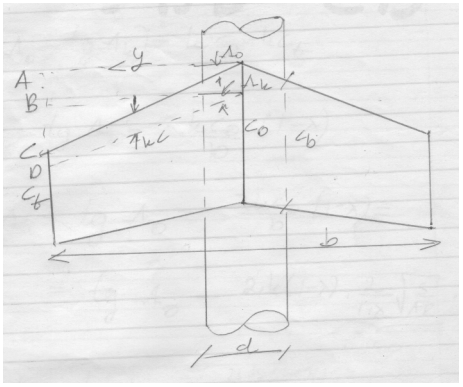
$S-wing\ area$

$AR-aspect\ ratio$

$\lambda=c_t/c_0$

$b-span$

$0\leqslant k\leqslant 1$



(i)

$$S=(c_0+c_t)/2*b$$

$$AR=b^2/S$$

$$S=c_0(1+\lambda)b/2=c_0(1+\lambda)/2*\sqrt{AR*S}$$

Hence

$$c_0=2/(1+\lambda)*\sqrt{(S/AR)}$$

(ii)

$$BC=AC-AB=BD-CD$$

$$(b/2)(\tan\Lambda_0-\tan\Lambda_k)=k\,c_0-k\,c_t$$

$$(b/2)(\tan\Lambda_0-\tan\Lambda_k)=k\,c_0(1-c_t/c_0)=k\,c_0(1-\lambda)$$

Rearrange

$$\tan\Lambda_k=\tan\Lambda_0-2k\,c_0(1-\lambda)/b$$

Subsitiute

$$c_0=2/(1+\lambda)*\sqrt{(S/AR)}$$

$$b=\sqrt{AR*S}$$

And we get

$$\tan\Lambda_k=\tan\Lambda_0-4k(1-\lambda)/[AR(1+\lambda)]$$

(iii)

c varies linearly

$$c=c_0+(c_t-c_0)/(b/2)*y$$

$$\bar{\bar{c}}=\frac{\int_0^{b/2}c^2\,dy}{\int_0^{b/2}c\,dy}$$

$$\int_0^{b/2}c\,dy=(c_t+c_0)b/4=c_0(1+\lambda)b/4$$

$$\int_0^{b/2}c^2\,dy=\int_0^{b/2}\left[c_0+(c_t-c_0)/(b/2)*y\right]^2dy$$

$$\int_0^{b/2}c^2\,dy=c_0^2*(b/2)+c_0(c_t-c_0)*(b/2)+(c_t-c_0)^2*(b/6)$$

$$\bar{\bar{c}}=\frac{2c_0\left[1+\lambda-1+(\lambda-1)^2/3\right]}{1+\lambda}=\frac{2c_0(3\lambda+\lambda^2-2\lambda+1)}{3(1+\lambda)}$$

$$\bar{\bar{c}}=\frac{2c_0(1+\lambda+\lambda^2)}{3(1+\lambda)}$$

(iv)

$$c = c_0 + (c_t - c_0) * (y/(b/2))$$

$$c_b = c(y = d/2) = c_0 + (c_t - c_0) * \frac{d/2}{b/2} = c_0 \left[1 + \frac{(\lambda - 1)d}{b} \right]$$

$$c_b = c_0 \left[1 - \frac{(1-\lambda)d}{b} \right]$$

2.2

Data:

$S = 0.25\,m^2$	$a_1 = 3.51/rad$	$b_1 = 0.031/rad$
$S_\delta = 0.06\,m^2$	$a_2 = 1.71/rad$	$b_2 = -0.0141/rad$
$\bar{\bar{c}}_\delta = 0.04\,m$	$a_3 = 0.31/rad$	$b_3 = -0.031/rad$

$\alpha_0 = 0$, $\alpha = 3°$, $\delta = -4°$, $\beta = 5°$, $b_0 = 0$

Solution:

(i)

Lift

$$C_L = a_1(\alpha_1 - \alpha_0) + a_2\delta + a_3\beta = (3.5 \cdot 3 + 1.7 \cdot (-4) + 0.3 \cdot 5) \cdot \pi/180 = 0.09$$

$$L = 1/2\,\rho U_\infty^2 S C_L = 0.5 \cdot 1.225 \cdot 60^2 \cdot 0.25 \cdot 0.09 = 49.6\,N$$

Torque

$$C_H = b_0 + b_1(\alpha_1 - \alpha_0) + b_2\delta + b_3\beta = 0 + (0.09 \cdot 3 + 0.014 \cdot 4 - 0.03 \cdot 5) \cdot \pi/180 = 3.072 * 10^{-3}$$

$$H = 1/2\,\rho U_\infty^2 S_\delta \bar{\bar{c}}_\delta C_H = 0.5 * 1.225 * 60^2 * 0.06 * 0.04 * 3.072 * 10^{-3} = 0.0163\,\text{N m}$$

(ii)

$$C_H = b_0 + b_1(\alpha_1 - \alpha_0) + b_2\delta + b_3\beta = 0 + (0.09 \cdot 3 - 0.014 \cdot \delta - 0.03 \cdot 5) \cdot \pi/180 = 0$$

$$\delta = 8.57^0$$

(iii)

$$C_L = a_1(\alpha_1 - \alpha_0) + a_2\delta + a_3\beta = (3.5 \cdot 3 + 1.7 \cdot 8.57 + 0.3 \cdot 5) \cdot \pi/180 = 0.464$$

$$L = 1/2\,\rho U_\infty^2 S C_L = 0.5 \cdot 1.225 \cdot 60^2 \cdot 0.25 \cdot 0.464 = 255.6\,N$$

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